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Randomized Kernel Ridge Regression

Topics in Numerical Mathematics A

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April 10, 2026

Overview

1. Kernel Ridge Regression
2. Classical Approach
3. Low-Rank Nature of the Kernel Matrix
4. Randomized Approach
5. Results & Conclusion

Kernel Ridge Regression

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Goal: Approximate some unknown function $g : \mathbb{R}^m \to \mathbb{R}$

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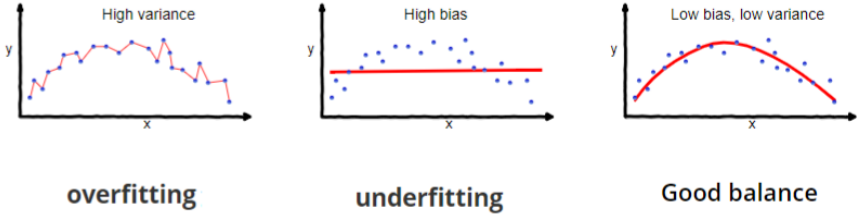
Minimization Problem

Find $\hat{f}$ such that:

$$\hat{f} = \arg\min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} (f(x_i) - y_i)^2$$

where $\mathcal{F}$ is the space of all well-defined functions mapping $\mathbb{R}^m \to \mathbb{R}$.

This approach is highly prone to over-fitting.



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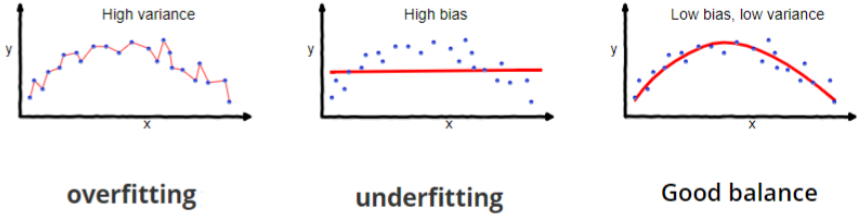

$\Longrightarrow$ add a regularization term to strike a balance.

$$\hat{f} = \underset{f \in \mathcal{F}}{\arg\min} \left(\frac{1}{n} \sum_{i=1}^{n} (f(x_i) - y_i)^2 + \lambda \|f\|_{\mathcal{F}}^2 \right), \qquad \lambda \in \mathbb{R}_{>0}$$

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How to choose the kernel function K for a given data set?

Commonly used kernel functions:

Linear kernel:	$K(x_1, x_2) = x_1^\top x_2$	
Polynomial kernel:	$K(x_1, x_2) = (1 + c x_1^\top x_2)^d$	with $c > 0,\ d \in \mathbb{N},\ (d = 2, 3)$
Gaussian (RBF) kernel:	$K(x_1, x_2) = e^{-c\|x_1 - x_2\|_2^2}$	with $c > 0$
Laplacian kernel:	$K(x_1, x_2) = e^{-c\|x_1 - x_2\|_1}$	with $c > 0$

Find the function $\hat{f} \in \mathcal{F}$ such that

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Representer Theorem

The minimizer $\hat{f}$ must be of the form $\hat{f}(\cdot) = \sum_{i=1}^{n} \hat{\alpha}_i K(x_i, \cdot)$ for $\hat{\alpha} \in \mathbb{R}^n$.

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Find the vector $\hat{\alpha} \in \mathbb{R}^n$ such that

$$\hat{\alpha} = \arg\min_{\alpha \in \mathbb{R}^n} \left(\frac{1}{n} \|K\alpha - y\|_2^2 + \lambda \alpha^\top K \alpha \right)$$

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$$K := \begin{bmatrix} K(x_1, x_1) & K(x_1, x_2) & \dots & K(x_1, x_n) \\ K(x_2, x_1) & K(x_2, x_2) & \dots & K(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ K(x_n, x_1) & K(x_n, x_2) & \dots & K(x_n, x_n) \end{bmatrix}$$

Kernel Ridge Regression

Find $\hat{\alpha}$ such that

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This is equivalent to finding $\hat{\alpha} \in \mathbb{R}^n$ such that

$$(K + n\lambda I)\hat{\alpha} = y$$

Classical Approach

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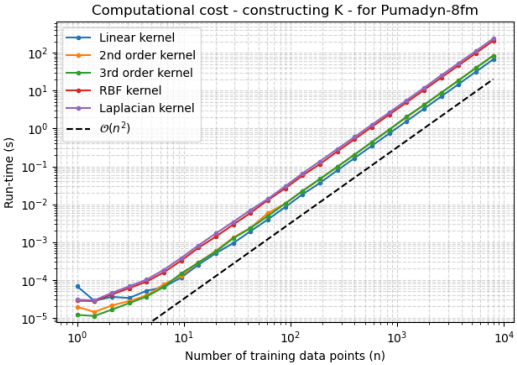
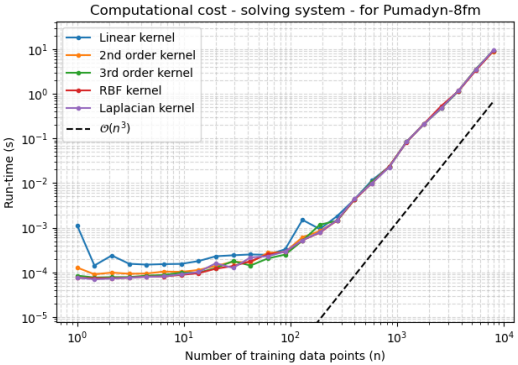
Computational cost:

- Constructing the kernel matrix K $\quad O(mn^2)$
- Solving the linear system $\quad O(n^3)$



Cost of computing K ($m = 8$)



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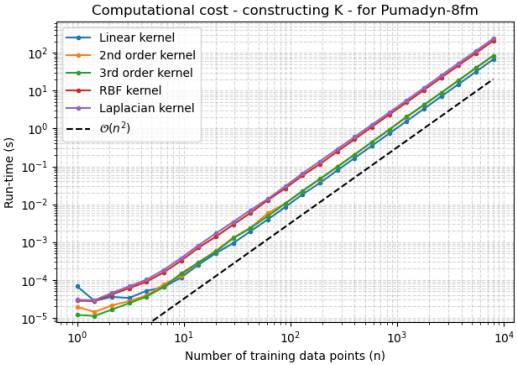

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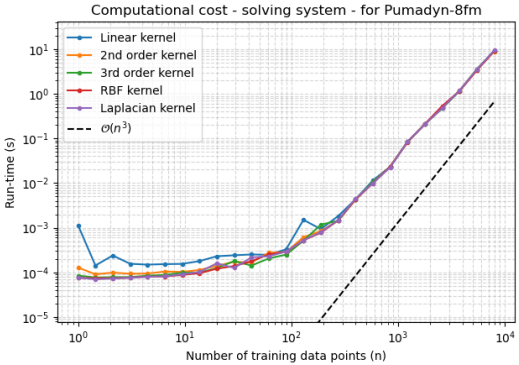

Cost of solving the system ($m = 8$)

Can we reduce the computational cost without compromising accuracy?

Low-Rank Nature of the Kernel Matrix

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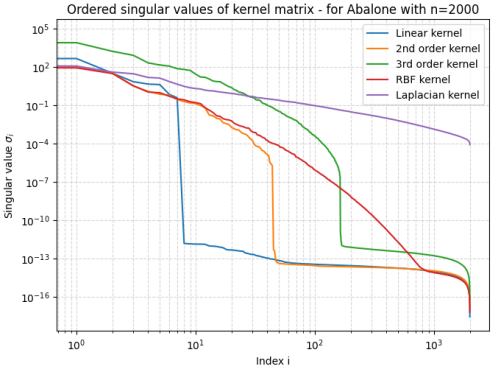
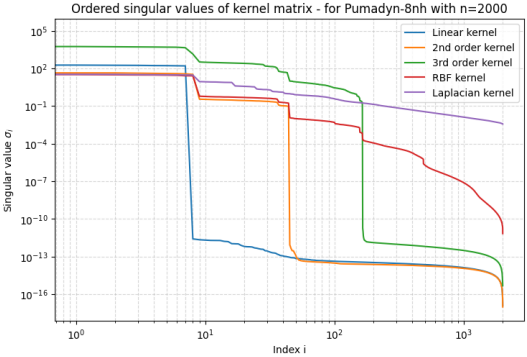
Singular Value Distribution of K

If we can show that the kernel matrix K has low rank, this will open the door to a choice of low rank approximations for K!

Ordered Singular Values



Singular value distribution on Abalone dataset



Singular value distribution on Pumadyn-8nh dataset

Randomized Approach

Nyström Approximation

Why Nyström?

- Preserves the symmetric PSD matrix structure

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- Preserves the symmetric PSD matrix structure
- Does not need the full construction of K
- Does not approximate the kernel function
- This preserves the data set's geometry
- Scalable with larger sampling sizes

Nyström Approximation

$K \in \mathbb{R}^{n \times n}$, $S \in \mathbb{R}^{n \times s}$ a sketching matrix with $s \ll n$ and columns of S are standard unit vectors.
The sampling of s columns of K can now be defined as

$$C := KS$$

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Definition

The Nyström Approximation of the kernel matrix K is

$$\hat{K} := C(S^\top KS)^\dagger R = CW^\dagger C^\top. \tag{1}$$

$$K = \left[\begin{array}{c|c|c|c} k_{11} & k_{12} & k_{13} & k_{14} \\ k_{12} & k_{22} & k_{23} & k_{24} \\ k_{13} & k_{23} & k_{33} & k_{34} \\ k_{14} & k_{24} & k_{34} & k_{44} \end{array}\right]$$

$$C = KS = \begin{bmatrix} k_{12} & k_{14} \\ k_{22} & k_{24} \\ k_{23} & k_{34} \\ k_{24} & k_{44} \end{bmatrix} \qquad R = S^\top K = \begin{bmatrix} k_{12} & k_{22} & k_{23} & k_{24} \\ k_{14} & k_{24} & k_{34} & k_{44} \end{bmatrix}$$

$$W = S^\top KS = \begin{bmatrix} k_{22} & k_{24} \\ k_{24} & k_{44} \end{bmatrix} \qquad \boxed{\hat{K} = C\,W^\dagger\,C^\top}^{[1]}$$

[1] Algorithm 5.6 (*NysSVD*) in Tropp and Webber (2023)

Solving $\hat{\alpha} = (\hat{K} + n\lambda I)^{-1} y$

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Woodbury Identity

$$(U\Lambda U^{\top} + n\lambda I)^{-1} = (n\lambda I)^{-1} - (n\lambda I)^{-1} U \left(\Lambda^{-1} + U^{\top}(n\lambda I)^{-1} U\right)^{-1} U^{\top}(n\lambda I)^{-1}$$

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Apply Woodbury Identity:

$$\begin{aligned}\hat{\alpha} &= (U\Lambda U^{\top} + n\lambda I)^{-1} y \\ &= \frac{1}{n\lambda}\left[y - U\,\mathrm{diag}\left(\frac{\Lambda_i}{n\lambda + \Lambda_i}\right) U^{\top} y\right]\end{aligned}$$

Column Sampling Approaches

1. **Uniform:** Choose columns of K uninformed

[2] Section 3.3 in Alaoui and Mahoney (2015)

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Column Sampling Approaches

1. **Uniform:** Choose columns of K uninformed
2. **Diagonal:** $p(K_i) = \frac{K_{ii}}{\sum_j K_{jj}}$
3. **Statistical Leverage Scores:** $\ell_i = ||(U_k)_i||_2^2, \quad p(K_i) = \frac{\ell_i}{\sum_j \ell_j}$

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4. λ**-Ridge LS:** $\ell_i(\lambda) = \sum_j \frac{\sigma_j}{\sigma_j + n\lambda} U_{ij}^2$

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4. λ**-Ridge LS:** $\ell_i(\lambda) = \sum_j \frac{\sigma_j}{\sigma_j + n\lambda} U_{ij}^2$
5. **Approx.** λ**-Ridge LS:** Uses a small set of sampled columns to build a fast low-rank $\hat{K}$, estimates scores from this sketch[2]

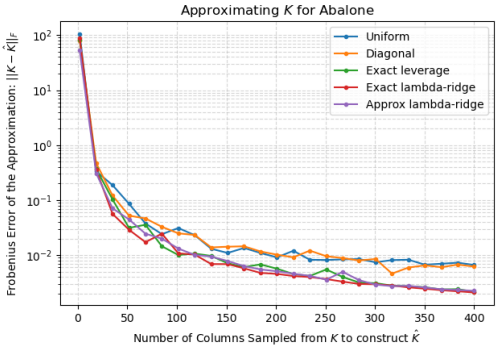
[2] Section 3.3 in Alaoui and Mahoney (2015)

Results & Conclusion

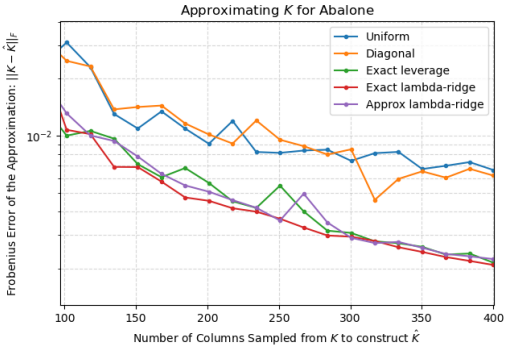
Approximating the kernel matrix K

- Approximating kernel matrix K using scoring methods
- Parameters: Abalone dataset, RBF kernel, $\lambda = 10^{-6}$, $n = 2000$, $n_{test} = 600$.
- Frobenius norm $||K - \hat{K}||_F$

Results kernel matrix K approximation



Frobenius norm $||K - \hat{K}||_F$ for Abalone



Frobenius norm $||K - \hat{K}||_F$ for Abalone

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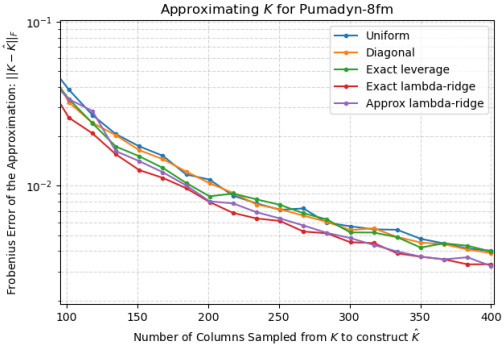
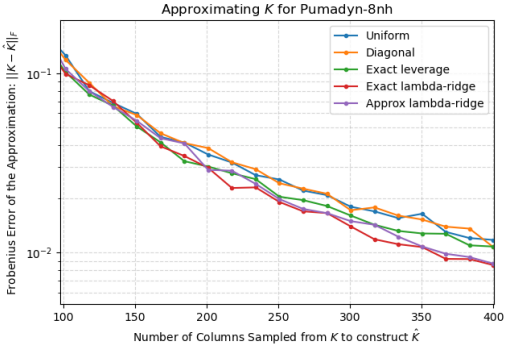


Frobenius norm $||K - \hat{K}||_F$ for Pumadyn-8fm

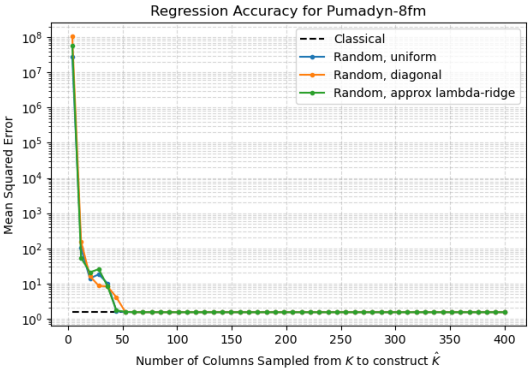


Frobenius norm $||K - \hat{K}||_F$ for Pumadyn-8nh

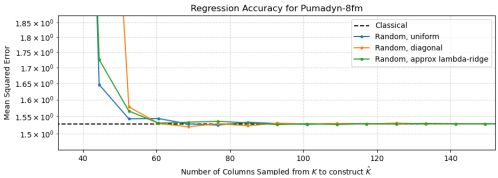
Results low-rank approximation vs. classical approach

- Column sampling with: uniform, diagonal and approximate λ-ridge
- Testing performance (run-time) and accuracy (MSE)
- Parameters: Pumadyn-8fm dataset, 2nd-order polynomial kernel, $\lambda = 10^{-5}$, $n = 2000$, $n_{test} = 600$.
- Comparison with classical approach

Results low-rank approximation vs. classical approach - MSE

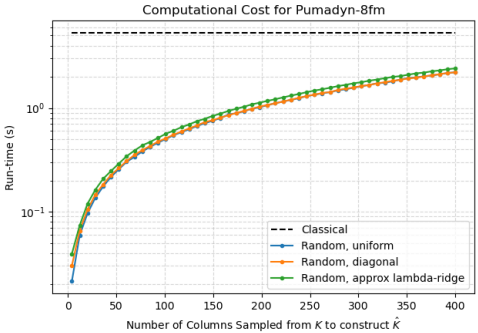


MSE for Pumadyn-8fm



MSE for Pumadyn-8fm

Results low-rank approximation vs. classical approach - Computational Cost



Computational cost for Pumadyn-8fm

Conclusion

- Does Nyström approximation with randomly sampled columns provide a good alternative to the classical approach?
- Should we use informed sampling for column selection?

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Thank you for your attention

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Data Sets

Each column is a feature, instances are rows and the target class is the final column

Abalone (Sea Snail)

- 8 features and 4177 instances
- Target Class - **number of rings** (integer) that appear on the abalone's shell
- Features - Continuous data: length, diameter, shell mass etc.
- First feature - sex (M, F or Infant) $\rightarrow$ (0, 1, 2)

Pumadyn

Pumadyn-8fm & **Pumadyn-8nh**

- Puma 560 robot arm (Puma Dynamics) - 8 data sets
- 8 or 32 features and 8192 instances
- Target Class - angular acceleration of robot arm's links
- Features - Continuous data: angular positions, velocities, torques etc.

NysSVD: Alg. 5.6 from Tropp & Webber (2023)

Inputs:

- $C \in \mathbb{R}^{n \times s}$, $W \in \mathbb{R}^{s \times s}$, $S \in \mathbb{R}^{n \times s}$, $\varepsilon_{\text{stab}}$

1. Perform stabilised Cholesky decomposition of W: $H^\top H = W + \varepsilon_{\text{stab}} S^\top S$
   - Since columns are sampled without replacement, $S^\top S = I_s$
2. Compute $Z = (C + \varepsilon_{\text{stab}} S) H^{-1}$
   - Since H is upper triangular, computing H^{-1} is efficient
3. Compute the SVD of Z: $[U, \Sigma, _] = \text{svd_econ}(Z)$
   - this is a 'thin' SVD, only returning the relevant columns of U, which are orthogonal
4. Remove the stabilisation: $\Lambda = \max\{0, \Sigma^2 - \varepsilon_{\text{stab}} I\}$

Outputs:

- U, Λ such that $\hat{K} \approx U \Lambda U^\top$

Approx. λ-Ridge LS from Alaoui & Mahoney (2015)

Inputs:

- data set X, samp. distr. $P = (p_1, \ldots, p_n)$, sampling parameter $p \ll n$, $\lambda > 0$

1. Sample p indices $i_1, \ldots, i_p$ from $\{1, \ldots, n\}$ with replacement according to P
2. Form the sampled columns $K_{i_1}, \ldots, K_{i_p}$ of the kernel matrix using X
3. Construct matrices
   - $C = [K_{i_1}, \ldots, K_{i_p}] \in \mathbb{R}^{n \times p}$
   - $W \in \mathbb{R}^{p \times p}$, where $W_{ab} = K(x_{i_a}, x_{i_b})$
4. Compute the factor B such that $BB^\top = CW^\dagger C^\top$
5. for each $i \in \{1, \ldots, n\}$, compute $\tilde{\ell}_i = B_i^\top (B^\top B + n\lambda I)^{-1} B_i$, where B_i is the i-th row of B

Outputs:

- list of values of $\tilde{\ell}_i$ for $i \in \{1, \ldots, n\}$